IV SEMESTER	L	T	P	C
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20MA4T07 : PROBABILITY AND STATISTICS				

COURSE OBJECTIVES:

1. To familiarize the students with the foundations of probability and statistical methods
2. To impart probability concepts and statistical methods in various applications of Engineering
3. To introduce the correlation and regression and method of least squares.

COURSE OUTCOMES:

At the end of this unit, the student will be able to

CO1: Make use of the concepts of probability and their applications (k3)
CO2: Apply discrete and continuous probability distributions(K3)
CO3: Design the components of a classical hypotheses test(K6)
CO4: Infer the statistical inferential metods based on small and large sampling tests(K6)
CO5: Adopt correlation methods and principle of least squares, regression lines (K_4)

UNIT-1: Probability:

Probability, probability axioms, addition law and multiplicative law of probability, conditional probability, Baye's theorem, random variables (discrete and continuous), probability density functions, properties, mathematical expectation.

Learning Outcomes:

At the end of this unit, the student will be able to

- define the terms trial, events, sample space ,probability, and laws of probability (K_1)
- make use of probabilities of events in finite sample space from experiments (K_3)
- apply Baye's theorem to real time problems(K_3)
- explain the notion of random variable, distribution functions and expected value(K_2)

UNIT-2: Probability distributions:

Probability distribution-Binomial, Poisson approximation to the binomial distribution and normal distribution –their properties.

Learning Outcomes:

At the end of this unit, the student will be able to

- apply Binomial and Poisson distributions for real data to compute probabilities, theoretical frequencies (K_3)
- interpret the properties of normal distribution and its applications (K_2)

UNIT-3: Sampling distribution and Testing of hypothesis, large sample tests:

Basic terminology in sampling, sample techniques (with and without replacement), sampling distribution of means for large and small samples (with known and unknown variance).

Formulation of null hypothesis, alternative hypothesis, the critical and acceptance regions, level of significance, two types of errors.

Large Sample Tests: Test for single proportion, difference of proportions, test for single mean and difference of means. Confidence interval for parameters in one sample and two sample problems

Learning Outcomes:

At the end of this unit, the student will be able to

- Explain the concept of sampling distribution for large and small samples (K_2)
- apply the concept of hypothesis testing for large samples (K_4)

UNIT-4: Small sample tests:

Student t-distribution (test for single mean, two means and paired t-test), testing of equality of variance (F-test),Chi-square test for goodness of fit and independence of attributes.

Learning Outcomes:

At the end of this unit, the student will be able to

- apply the concept of testing of hypothesis for small samples to draw the inferences (K_3)
- estimate the goodness of fit (K_4)

UNIT-5: Curve Fitting and Correlation:

Curve Fitting : Method of least squares -Fitting a straight line, Second degree parabola -exponential curve-power curves

Correlation: Simple correlation, correlation coefficient (for ungrouped data), rank correlation. Linear regression, regression lines, regression coefficients.

Learning Outcomes:

At the end of this unit, the student will be able to

- Adopt correlation methods and principle of least squares, regression lines (K_4)

Text Books:

1. Probability and Statistics by Dr. K. Murugesan &P.Gurusamy, Anuradha Publications,2011
2. Higher Engineering Mathematics, 42nd Edition, Khanna Publications by Dr. B.S.Grewal,2012

Reference Books:

1. Ramana B.V., Higher Engineering Mathematics, Tata Mc Graw Hill New Delhi 11th Reprint2010
2. Probability and statistics for engineers by Miller & Freund